

1. Employees Above Their Department Average Display emp_id, emp_name, dept_id, and salary for employees whose salary is greater than the average salary of their own department. Do not use JOIN. Focus: correlated subquery with AVG().

ANS: select e.emp_id,e.emp_name,e.dept_id,e.salary from employees e where e.salary > (select avg(salary) from employees where dept_id=e.dept_id);

2. Highest Paid Employee(s) in Every Department Display the highest-paid employee or employees from each department. If multiple employees in the same department share the highest salary, return all of them. Focus: correlated subquery with MAX().

ANS: select e.emp_id,e.emp_name,e.dept_id,e.salary from employees e where salary = (select max(salary) from employees where dept_id=e.dept_id);

3. Second Highest Distinct Salary Find the second-highest distinct salary in the company without using LIMIT, OFFSET, or JOIN. Focus: nested scalar subquery.

ANS: select max(salary) from employees where salary < (select max(salary) from employees);

4. Employees Receiving the Second Highest Distinct Salary Display the employee name and salary of every employee whose salary equals the second-highest distinct salary in the company. Focus: nested subqueries and duplicate-safe result.

ANS: select emp_id,emp_name,salary from employees where salary = (select max(salary) from employees where salary < (select max(salary) from employees));

5. Employees Earning More Than Their Manager Display employees whose salary is greater than the salary of their own manager. Employees without a manager should not appear. Do not use a self JOIN. Focus: correlated scalar subquery against the same table.

ANS: select emp_id,emp_name,salary from employees e where salary > (select salary from employees where emp_id=e.manager_id);

6. Managers Who Earn Less Than At Least One Direct Report Display employees who are managers and whose salary is less than at least one employee reporting directly to them. Focus: correlated subquery with EXISTS or ANY.

ANS: select emp_id,emp_name,salary from employees e where salary < any (select salary from employees where manager_id=e.emp_id);

7. Departments Whose Average Salary Is Above Company Average Display the department IDs whose average employee salary is greater than the overall average salary of all employees. Focus: aggregate query with a scalar subquery in HAVING.

ANS: select distinct dept_id from employees group by dept_id having avg(salary) > (select avg(salary) from employees);

8. Employees in Departments Whose Average Salary Is Above Company Average Display all employees who belong to departments whose departmental average salary is greater than the overall company average salary. Do not use JOIN. Focus: nested multiple-row subquery.

ANS: select emp_id,emp_name,dept_id from employees where dept_id in (select dept_id from employees group by dept_id having avg(salary)> (select avg(salary) from employees));

9. Employees Earning More Than Every Employee in HR Display employees whose salary is greater than the salary of every employee in the HR department. The department ID must be obtained using a subquery from the departments table. Focus: nested subquery with ALL.

ANS: select emp_id, emp_name, salary from employees where salary > all(select salary from employees d where dept_id in (select dept_id from departments where dept_name='HR'));

10. Employees Earning More Than At Least One Research Employee Display employees whose salary is greater than at least one employee working in the Research department. Obtain the Research department ID through a subquery. Focus: nested subquery with ANY.

ANS: select emp_id, emp_name, salary from employees where salary > any(select salary from employees d where dept_id in (select dept_id from departments where dept_name='Research'));

11. Departments Having No Projects Display dept_id and dept_name for departments that do not have any project listed in the projects table. Solve using a subquery only. Focus: NOT EXISTS correlated subquery.

ANS: select dept_id, dept_name from departments d where dept_id in(select dept_id from projects where dept_id = d.dept_id);

12. Employees Belonging to Departments With More Than One Project Display all employees who belong to a department that has more than one project. Do not use JOIN. Focus: correlated / grouped subquery with COUNT().

ANS: select emp_id, emp_name, dept_id from employees where dept_id in (select dept_id from projects group by dept_id having count(project_id)>1);

13. Employees in the Department of the Highest-Budget Project Display all employees who work in the department that owns the project having the maximum budget. If multiple projects tie for maximum budget, employees from all matching departments should be returned. Focus: nested multi-row subquery with MAX().

ANS: select emp_id, emp_name from employees where dept_id in (select dept_id from projects where budget = (select max(budget) from projects));

14. Employee(s) With Salary Closest to Company Average Display the employee or employees whose salary has the smallest absolute difference from the company average salary. Do not use ORDER BY ... LIMIT and do not use JOIN. Focus: nested aggregate subqueries using AVG(), ABS(), and MIN().

ANS: select emp_id, emp_name, salary from employees where abs(salary - (select avg(salary) from employees))= (select min(abs(salary-(select avg(salary) from employees))) from employees);

15. Departments Whose Top Salary Exceeds Every Manager Salary in That Department Display department IDs where the maximum employee salary in that department is greater than the salary of every employee in the same department who is acting as a manager of at least one other employee. Focus: advanced correlated and nested subqueries with MAX(), ALL / NOT EXISTS.

ANS: SELECT dept_id FROM employees e GROUP BY dept_id HAVING MAX(salary) > ALL (SELECT m.salary FROM employees m WHERE m.dept_id = e.dept_id AND m.emp_id IN (SELECT DISTINCT manager_id FROM employees WHERE manager_id IS NOT NULL));